

5.17 Classwork: Logarithms Practice — Markscheme

1. [Maximum mark: 12]

Evaluate each of the following.

Solution:

$$\text{(a)} \quad \frac{1}{64} = 4^{-3} \quad \text{M1}$$

$$\log_4 \frac{1}{64} = -3 \quad \text{A1}$$

$$\text{(b)} \quad 5 = 25^{1/2} \quad \text{M1}$$

$$\log_{25} 5 = \frac{1}{2} \quad \text{A1}$$

$$\text{(c)} \quad \sqrt{32} = 2^{5/2} \quad \text{M1}$$

$$\log_2 \sqrt{32} = \frac{5}{2} \quad \text{A1}$$

$$\text{(d)} \quad 125 = 5^3 \quad \text{M1}$$

$$\log_5 125 = 3 \quad \text{A1}$$

$$\text{(e)} \quad \frac{1}{4} = 16^{-1/2} \quad \text{M1}$$

$$\log_{16} \frac{1}{4} = -\frac{1}{2} \quad \text{A1}$$

$$\text{(f)} \quad \sqrt{81} = 9 = 3^2 \quad \text{M1}$$

$$\log_3 \sqrt{81} = 2 \quad \text{A1}$$

2. [Maximum mark: 12]

(a) Let $c = \log_3 m$, where $m > 0$. Write each expression in terms of c . [6]

Solution:

$$\begin{aligned} \text{(i)} \quad \log_3 m^2 &= 2 \log_3 m && \text{M1} \\ &= 2c && \text{A1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \log_3(27m) &= \log_3 27 + \log_3 m && \text{M1} \\ &= 3 + c && \text{A1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \log_9 m &= \frac{\log_3 m}{\log_3 9} && \text{M1} \\ &= \frac{c}{2} && \text{A1} \end{aligned}$$

(b) Let $a = \log_2 m$, where $m > 0$. Write each expression in terms of a . [6]

Solution:

$$\begin{aligned} \text{(i)} \quad \log_2 m^3 &= 3 \log_2 m && \text{M1} \\ &= 3a && \text{A1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \log_2(8m) &= \log_2 8 + \log_2 m && \text{M1} \\ &= 3 + a && \text{A1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \log_4 m &= \frac{\log_2 m}{\log_2 4} && \text{M1} \\ &= \frac{a}{2} && \text{A1} \end{aligned}$$

3. [Maximum mark: 12]

Use the laws of logarithms.

(a) Express each in terms of $\log x$ and $\log y$. [6]

Solution:

$$(i) \log(100x) = \log 100 + \log x \quad \mathbf{M1}$$

$$= 2 + \log x \quad \mathbf{A1}$$

$$(ii) \log\left(\frac{x^2}{y}\right) = \log x^2 - \log y \quad \mathbf{M1}$$

$$= 2 \log x - \log y \quad \mathbf{A1}$$

$$(iii) \log \sqrt{xy^3} = \frac{1}{2} \log(xy^3) \quad \mathbf{M1}$$

$$= \frac{1}{2} \log x + \frac{3}{2} \log y \quad \mathbf{A1}$$

(b) Write each expression as a single logarithm. [6]

Solution:

$$(i) \log_2 x + \log_2 y = \log_2(xy) \quad \mathbf{M1A1}$$

$$(ii) 2 \log_7 a - \log_7 b = \log_7 a^2 - \log_7 b \quad \mathbf{M1}$$

$$= \log_7\left(\frac{a^2}{b}\right) \quad \mathbf{A1}$$

$$(iii) \frac{1}{2} \ln p + \ln q = \ln p^{1/2} + \ln q \quad \mathbf{M1}$$

$$= \ln(q\sqrt{p}) \quad \mathbf{A1}$$

Accept $\ln(\sqrt{p}q)$ or $\ln(p^{1/2}q)$.

4. [Maximum mark: 12]

Solve each equation. Give exact answers.

Solution:

$$\text{(a)} \log_4(x+9) = \frac{1}{2} \log_2(x+9) \quad \text{M1}$$

$$2 \log_2(x-3) = \log_2(x+9)$$

$$(x-3)^2 = x+9 \quad \text{M1}$$

$$x^2 - 7x = 0$$

$$x = 0 \text{ or } x = 7$$

$$\text{Domain requires } x > 3, \text{ so } x = 7 \quad \text{A1}$$

$$\text{(b)} \log_3 x = 2$$

$$x = 3^2 \quad \text{M1}$$

$$x = 9 \quad \text{A1A1}$$

$$\text{(c)} \log_5(2x) = 1$$

$$2x = 5^1 \quad \text{M1}$$

$$x = \frac{5}{2} \quad \text{A1A1}$$

$$\text{(d)} \log_2(x+1) + \log_2(x-1) = \log_2((x+1)(x-1)) \quad \text{M1}$$

$$x^2 - 1 = 8 \quad \text{M1}$$

$$x^2 = 9$$

$$x = \pm 3$$

Reject $x = -3$ since $x > 1$.

$$x = 3 \quad \text{A1}$$